

Technical Appendix

Case GRIDSHIELD – Decode X 2026 | Case 02

Model Specifications

Model Type: XGBoost Regressor

Objective: Minimize asymmetric financial deviation penalty under ABT framework

Forecast Resolution: 15-minute intervals

Forecast Horizon: 2-day ahead

Training Period: April 2013 – December 2019

Validation Period: January 2020 – December 2020

Test Period (Out-of-Time): January 2021 – April 2021

Target Variable: Load (kW)

Penalty Structure Applied:

- Stage 1:
 - Under-forecast → ₹4 per kWh
 - Over-forecast → ₹2 per kWh
- Stage 2 & 3:
 - Peak Under-forecast (6 PM – 10 PM) → ₹6 per kWh
 - Off-peak Under-forecast → ₹4 per kWh
 - Over-forecast → ₹2 per kWh

Model Configuration (Conceptual):

- Gradient Boosted Decision Trees
- Learning rate controlled to avoid overfitting
- Tree depth optimized via validation
- Early stopping based on validation error
- Chronological data split (no shuffling)

Feature Engineering

Time-Based Features

- Hour of day
- Day of week
- Month
- Day of year
- Weekend indicator
- Peak-hour indicator (6 PM–10 PM)
- Cyclical encoding (sin/cos transformation of hour)

Lag Features

- 1 interval lag (15 minutes)
- 4 interval lag (1 hour)
- 96 interval lag (1 day)
- 672 interval lag (7 days)

Rolling Statistics

- 96-interval rolling mean (24-hour smoothing)
- 16-interval rolling mean (4-hour trend capture)
- 96-interval rolling standard deviation (volatility capture)

External Drivers

- Temperature
- Humidity

- Heat Index
- Rainfall
- Cool Factor
- Event / Holiday Indicator

All rolling and lag features were computed using past-only windows to prevent leakage.

Assumptions

1. Weather variables reflect realistic 2-day ahead forecast conditions.
 2. Structural disruptions (e.g., pandemic impact) are embedded in historical data.
 3. No perfect stationarity assumed across years.
 4. Demand elasticity may shift but remains within historical volatility bounds.
 5. Revised penalty structure is binding and must be incorporated directly into optimization logic.
 6. Procurement inefficiency risk exists if upward bias exceeds regulatory tolerance.
 7. All timestamps are aligned to 15-minute resolution without gaps.
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Methodology

Stage 1 – Baseline Cost-Aware Forecasting

- Built XGBoost regression model.
- Optimized hyperparameters on validation year (2020).
- Introduced upward bias (+3 units) to reduce expensive under-forecast exposure.
- Evaluated financial penalty instead of RMSE alone.

Stage 2 – Regime Shift Adaptation

- Applied trained model to out-of-time 2021 dataset.
- Incorporated revised peak penalty (₹6 under-forecast).
- Quantified exposure under new asymmetry.
- Introduced peak-aware bias recalibration using structured bias search.
- Reduced high-cost peak exposure significantly.

Stage 3 – Constrained Optimization

Reformulated objective:

Minimize total financial penalty

Subject to:

- Total exposure cap
- Peak reliability constraint
- Bias bound (-2% to +3%)
- Buffering constraint ($\leq 3\%$ uplift)

Implemented moderate peak-aware bias strategy to balance financial prudence and reliability.

Validation Approach

Time-Aware Validation

- Strict chronological split.
- No random sampling.
- Training: 2013–2019
- Validation: 2020
- Test: 2021

Backtesting Metrics

- Total deviation penalty (₹)
- Peak-hour penalty (₹)
- Off-peak penalty (₹)
- Forecast bias (%)
- 95th percentile absolute deviation (kW)

Leakage Controls

- Lag features use only historical data.
- Rolling statistics computed with backward windows.
- No test data used during model tuning.
- No future weather leakage assumed.

Stage 1 Backtest Metrics

Validation Year (2020):

- Controlled positive bias ($\sim +0.35\%$)
- Significant reduction vs seasonal naive benchmark
- 95th percentile deviation maintained within acceptable band
- Peak-hour exposure monitored

Bias introduction justified due to asymmetric 2:1 cost structure.

Stage 2 Impact Quantification

Under revised peak penalty:

Baseline (No Recalibration):

- Total penalty materially increased.
- Peak-hour exposure sharply elevated.

After Peak-Aware Recalibration:

- ~60% reduction in total financial exposure.
- ~75% reduction in peak penalty.
- Risk redistributed away from peak concentration.

Structural insight:

Optimal forecast quantile shifted upward under 3:1 asymmetry.

Stage 3 Constraint Verification

Final solution verified against all constraints:

- Total penalty maintained below Board cap.
- Peak underestimation >5% limited to ≤ 3 intervals.
- Forecast bias maintained within [-2%, +3%].
- Average forecast uplift maintained $\leq 3\%$.
- Full penalty breakdown reported (Total, Peak, Off-Peak).

Balanced bias positioning achieved without violating regulatory discipline.

Sensitivity Analysis

1. Penalty vs Peak Bias Curve

- Increasing peak bias reduces under-forecast cost.
- Excess bias increases over-procurement risk.

- Optimal region identified around moderate bias.

2. Bias vs Uplift Relationship

- Aggressive uplift violates buffering constraint.
- Controlled uplift preserves procurement discipline.

3. Peak Volatility Stress Test

- Simulated $\pm 10\%$ peak demand amplification.
- Strategy remains stable under moderate volatility shift.

4. Asymmetry Escalation Scenario

- If penalty asymmetry increases further, optimal quantile shifts higher.
- Framework adaptable to revised regulatory parameters.

Limitations & Model Governance

- Extreme demand shocks beyond historical range may increase deviation exposure.
- Weather forecast error may affect peak intervals.
- Regulatory asymmetry beyond current structure may require recalibration.
- Model performance dependent on timely weather data availability.

Governance Framework:

- Continuous penalty monitoring dashboard.
- Rolling volatility tracking.
- Periodic bias recalibration review.
- Transparent reporting of deviation exposure metrics.